Meta Context Engineering via Agentic Skill Evolution

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HarnessRSI 63 ★★

evidence: improver

Paper

1 What changes, what is kept

Skills = folders, not model weights
Instructions + scripts + templates + operators
All underlying LLM weights remain frozen

1. Skill history → new skill
Meta-agent: MiniMax M2.1
Reads skills, executions, train/val scores; combines and refines earlier procedures.

2. Execute the skill
Base-agent warm-starts best context.
Learns from labelled training rollouts; edits context files and retrieval code.

3. Evaluate the candidate
Training metrics + validation metrics.
Task-label accuracy / exact match / F1; no LLM grading judge specified.

4. Keep best on validation
Archive every trial; retain better context.
Return best context + corresponding skill.
Tie-breaking is not specified.

Algorithm 1 starts with empty history/context. Later iterations warm-start the prior best. Five training epochs; one-shot query → context → task-model inference.

Offline: validation selects; held-out test evaluates. Online: fixed/no skill; no iterative skill evolution.

2 Offline: best in each column

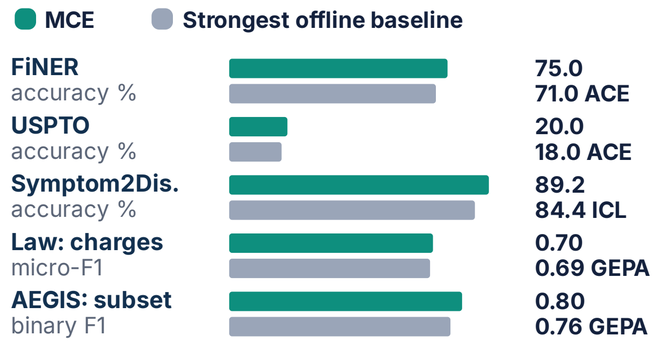


Table 1. Law: criminal charge prediction. AEGIS: selected-category binary safety. USPTO: product SMILES + reaction type supplied. Accuracy 0–100; F1 0–1. No seeds/SD/CI reported.

89.1% average relative gain is over base, not best baseline (ACE: 70.7%). Our read: USPTO's low base (6.0%) weighs heavily; final accuracy is only 20.0%.

4 Where the gain is thin or absent

- LawBench offline: **0.70 vs GEPA 0.69**; no uncertainty estimate.
- Online fixed skill loses to no skill on **Symptom 76.4 vs 76.9; Law 0.66 vs 0.70; AEGIS 0.63 vs 0.68**. Benefits concentrate on USPTO (20.0 vs 18.0); FiNER 68.0 vs 67.0.
- Context transfer to Gemma: FiNER **65.0 vs ACE 64.0**; Law **0.03 vs 0.00**, still near zero. Not cross-task skill transfer.

3 Controls, costs and boundaries

FiNER ablation · Table 3 · accuracy %

Method	Offline	Online
Evolving skill	**75.0**	not used
No skill	73.0	67.0
Fixed skill	71.0	68.0
ACE	71.0	64.0

Ablation only on FiNER; no variance given.

Baseline access: same starting templates, data/order, labels when applicable. ICL: fitting demos; MIPROv2/GEPA: DSPy heavy; GEPA: matched rollouts; DC: cumulative; ACE: original settings, five epochs or validation/window stop.

Model control: ACE with MiniMax: 67 vs DeepSeek 70 (final). No baseline given MCE's skill/file optimizer.

FiNER training cost: 1.9 h vs ACE 25.8 h; 450 rollouts to 95% train accuracy vs ACE peak 94% after 2169. MCE includes train + validation inference. No full dollar/token accounting.

Split limits: validation selects context; default/hyperparameter selection split and tuning-search budgets unspecified. Missing numeric generator decoding settings, batch schedule and ACE plateau threshold.

Leakage: random/subset splits; symptom train/test descriptions are close. Skills can encode examples and keyword rules. Permissions are described; no empirical duplicate or held-out-label audit.

Takeaway

FiNER supports evolving the context-building procedure; online fixed skills can hurt. Authors caution: reasoning-intensive tasks may not benefit; MCE may struggle with long, complex trajectories needing fine-grained credit assignment. These are cautions, not measured failures.

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Reports: holdout, rollout match, skill ablation, transfer, cost.
Missing: seeds, leakage audit, cross-task skills, compute parity.